

Data Vortex · Phase 2 challenge set

Easy, Medium and Hard reasoning challenges — answered in SQL against the restored Social Engine database.

#	Challenge	Answer	Why it can be trusted
E1	Highest-volume platform	YouTube · 1,770 posts	field is statistically flat (χ^2 2.61 < 9.49)
E2	Top-10 most engaged posts	7,893 at #1	likes required; 1,532 posts excluded
E3	Best average engagement	Instagram · 3,673.2	gap not significant (z 0.64)
E4	Shared hard, liked little	214 posts	shares > 1,500 and likes < 500
E5	Audiences over 40k	293 users	strictly greater than 40,000
M1	Engagement by location	Los Angeles, USA	rank follows post volume, not quality
M2	Followers vs engagement	no advantage	3,598.8 high vs 3,648.0 low
M3	Most active users	17 posts at #1	12 users tie on the 10th-place count
M4	Best platform for 30k+ accounts	Instagram	same winner at every threshold 25k-45k
M5	Over-shared posts	20 shown	1047 in total; 1154 blank-likes rows excluded
H1	2x average engagement	no user qualifies	best is 1.81x of the baseline
H2	Top 3 per location	99 users	smallest podium margin 0.08%
H3	Exceptional posts per platform	56 posts	each judged against its own platform mean
H4	Small accounts in the top decile	17 users	two percentile definitions agree
H5	Corrupted posts	4418 posts	4 defect families, scanned on the raw intake
H6	Suspicious high-impact users	82 users	all three conditions, from a 1,500-user funnel

Every number in this document was executed, not typed

The tables below are rendered from output/phase2_sql_results.json, which src/run_sql.py writes by running queries/phase2_challenges.sql against output/social_engine.db. Re-run ./run_all.sh and you land on identical figures. The companion queries are part of the answer: each headline result is attacked by its own assumptions before it is published.

How to read this report

Five rules govern every query in the set, and each one exists because breaking it would have produced a confident wrong answer somewhere in this document.

- **R1 · Engagement** is `COALESCE(likes,0)+COALESCE(shares,0)+COALESCE(comments,0)`. SQL's bare `+` returns `NULL` if ANY operand is `NULL`, so a naive `SUM` would silently drop 1,532 rows (15%) from every aggregate instead of raising an error.
- **R2 · Nothing is imputed.** 1,532 posts have no recorded like count. Where a challenge says to ignore them (E2) or where the comparison is undefined without them (M5, H6) the query filters them out. Reading a blank as zero would have invented 1,154 M5 'anomalies' out of missing data alone.
- **R3 · 'Unspecified' is a data gap**, not a platform (1,541 posts, 15.08%). It is kept for volume questions, excluded wherever a per-platform rate or a winner is computed, and every exclusion is priced by a companion query.
- **R4 · Ordering is total.** Every `LIMIT` is paired with a deterministic tie-break, because an arbitrary tie-break is how a 'top 10' becomes a different list on the next run.
- **R5 · Two readings? Both are priced.** 17 companion queries (E1b, E2b, E3b, E4b, M1b, M2b, M3b, M4b, M5b, H1b, H1c, H2b, H3b, H4b, H5b, H5c, H6b) quantify the alternative interpretation rather than asserting the chosen one.

Data: 10,221 analysis posts and 1,500 users, plus the 12,360-row raw intake file staged for the corruption questions.

Easy challenges

E1 · Platform Popularity

Challenge. Determine which social media platform has the highest number of posts. Ignore posts where the platform is missing. Return the platform name and number of posts.

platform	n_posts
YouTube	1770

Answer: YouTube with 1,770 posts. The two named columns are exactly what the challenge asks for; the 1,541 posts whose platform label was lost are excluded before the ranking, so a data gap cannot top it.

E1b · Is the leader actually ahead?

The gap between YouTube (1,770) and Instagram (1,689) is 81 posts, which sounds decisive until it is tested. Against a uniform expectation of 1,736.0 posts per platform, the chi-square statistic is **2.6094** with 4 degrees of freedom — well inside the 9.488 critical value at 5%. The five labelled platforms are statistically indistinguishable, so the honest finding is **a tie broken by noise**, not a platform preference. Publishing 'YouTube dominates' from the raw ranking would have been the easiest mistake in this set to make.

platform	n_posts	pct_shares	expected_if_uniform	chi2_term	chi2_total	df	verdict_at_5pct
YouTube	1770	20.39	1736.0	0.6659	2.6094	4	field is flat -- the leader is ahead only by noise
Facebook	1763	20.31	1736.0	0.4199	2.6094	4	field is flat -- the leader is ahead only by noise
Twitter	1742	20.07	1736.0	0.0207	2.6094	4	field is flat -- the leader is ahead only by noise
Reddit	1716	19.77	1736.0	0.2304	2.6094	4	field is flat -- the leader is ahead only by noise
Instagram	1689	19.46	1736.0	1.2725	2.6094	4	field is flat -- the leader is ahead only by noise

E2 · Most Engaged Posts

Challenge. Find the top 10 posts based on total engagement. Total engagement is defined as likes + shares + comments. Ignore posts where likes are missing.

post_id	user_id	platform	likes	shares	comments	total_engagement
ycjj5zzt7mvx	user_d9971ba6	Instagram	4983	1919	991	7893
wo7py9alvg3t	user_o8le7hqf	Reddit	4864	1981	948	7793
gmoeib832zbs	user_pe5yckyby	Facebook	4902	1880	982	7764
5kvuyvf38nqx	user_z0feut2e	YouTube	4923	1971	861	7755

post_id	user_id	platform	likes	shares	comments	total_engagement
pvfl3d8hj7jd	user_csluibwk	Instagram	4989	1840	909	7738
tdgjyjlpu20	user_8nvzxsuj	Unspecified	4979	1932	812	7723
tne7s3o4l4wd	user_lr3fagd1	Instagram	4931	1903	878	7712
a1kiwl618kzy	user_aaiari8o	Facebook	4811	1952	920	7683

Answer: 10 posts, led by ycj5zzt7mvx (7,893 engagement). 1,532 posts are excluded because their like count is missing — ranking a post on incomplete evidence is what the challenge's own instruction prevents.

E2b · The cut is not a cliff. 8,689 posts are eligible; the leader scores 7,893 and tenth place scores 7,583 — only 4.09% apart, with 14 posts within 1% of the cut. The 'top 10' is a slice through a dense cloud, and the report says so rather than implying ten outliers.

E3 · Average Engagement By Platform

Challenge. Calculate the average likes, shares, and comments for each platform. Which platform generates the highest average total engagement?

platform	n_posts	rows_with_likes	avg_likes	avg_shares	avg_comments	avg_total_engagement	engagement_rank	eligibility
Instagram	1689	1435	2504.8	1043.3	501.8	3673.2	1	labelled platform
Unspecified	1541	1323	2501.2	999.2	500.4	3646.9	2	DATA GAP -- not a platform, not eligible to win
Facebook	1763	1494	2527.3	986.8	507.6	3636.1	3	labelled platform
YouTube	1770	1495	2494.9	1009.0	505.3	3621.5	4	labelled platform
Reddit	1716	1473	2463.8	985.1	506.9	3606.9	5	labelled platform
Twitter	1742	1469	2432.4	1006.8	503.5	3561.5	6	labelled platform

Answer: Instagram generates the highest average total engagement, 3,673.2 per post against 3,636.1 for Facebook. The averages are taken over the posts that actually have a like count, never over imputed zeros.

E3b · And the win is not statistically real. The 37.0-engagement gap is 1.02% of the runner-up mean and carries a z-statistic of 0.635 — under the 1.96 bar, so the per-platform ordering is a coin toss. Both statements belong in the answer: Instagram leads, and the lead is not evidence.

E4 · Highly Shared But Poorly Liked

Challenge. Identify posts that received more than 1,500 shares but fewer than 500 likes. Return the post ID, platform, likes, shares, and comments.

post_id	platform	likes	shares	comments
euvr0r10wrj6	Facebook	453	2000	408
oisz0jqm6qnn	Instagram	390	1999	858
f2e5kdfldedz	Unspecified	447	1997	641
mgv7p46wzpek	Reddit	252	1993	971
u1aa801qvxeu	Unspecified	162	1992	306
0nsga7zrxpvt	YouTube	390	1991	848
6s9mxoemn488	Twitter	169	1991	856
lit2hyqg0v0l	Facebook	14	1990	769

Answer: 214 posts satisfy shares > 1,500 AND likes < 500. Both bounds are strict, so no post sitting exactly on a boundary is included. A post with a missing like count cannot satisfy 'fewer than 500 likes' and is correctly absent.

E5 · Users With Large Audiences

Challenge. Find all users with more than 40,000 followers. Display their user ID, location, language, and follower count.

user_id	location	language	follower_count
user_3o7w66o2	Berlin, Germany	ja	49944

user_id	location	language	follower_count
user_u98jwp3f	Chicago, USA	zh	49936
user_usts5yuo	Shanghai, China	ja	49933
user_d4eat3v3	Tokyo, Japan	de	49914
user_siuvpkza	Chicago, USA	en	49905
user_ujllj1n7	Chicago, USA	es	49855
user_kr0ydzhh	Barcelona, Spain	fr	49836
user_9asov0m8	Singapore	hi	49763

Answer: 293 users have more than 40,000 followers. Location is the source's own string, kept verbatim in the users table, so every row is traceable to the intake file character-for-character.

E4b · Is the over-sharing pattern platform-specific?

Counts alone would just re-report platform size, so the companion query reports the **rate**: flagged posts as a share of that platform's own posts. The pattern is densest on Instagram (2.61% of its posts) and it appears on every platform, so this is a corpus-wide behaviour rather than one platform's quirk.

platform	n_posts	flagged_posts	pct_of_platform
Instagram	1689	44	2.61
Facebook	1763	42	2.38
Twitter	1742	39	2.24
Unspecified	1541	31	2.01
YouTube	1770	35	1.98
Reddit	1716	23	1.34

Medium challenges

M1 · Engagement By Location

Challenge. Using both datasets, calculate the total engagement generated by users from each location. Return the location, number of posts, and total engagement. Rank locations from highest to lowest engagement.

location	n_posts	total_engagement	engagement_rank
Los Angeles, USA	391	1468866	1
Munich, Germany	388	1423109	2
Barcelona, Spain	376	1384741	3
Houston, USA	369	1331509	4
Dubai, UAE	360	1316970	5
Shanghai, China	374	1301061	6
Melbourne, Australia	358	1296792	7
Osaka, Japan	342	1253204	8

Answer: 33 locations, led by Los Angeles, USA with 1,468,866 total engagement across 391 posts. This is the one challenge that genuinely needs both datasets: engagement lives on the post, location lives on the user.

M1b · The ranking is about volume, not quality

Total engagement is volume × quality, so the two are ranked separately. Shanghai, China is 6th by total but 33rd by engagement per post — a shift of 27 places. Reporting the total ranking as 'the most engaged locations' would have published a post-count artefact as an audience insight.

location	n_posts	total_engagement	avg_engagement_per_post	rank_by_total	rank_by_avg_per_post	rank_shift
Los Angeles, USA	391	1468866	3756.7	1	1	0
Munich, Germany	388	1423109	3667.8	2	9	-7
Barcelona, Spain	376	1384741	3682.8	3	7	-4
Houston, USA	369	1331509	3608.4	4	19	-15
Dubai, UAE	360	1316970	3658.3	5	11	-6
Shanghai, China	374	1301061	3478.8	6	33	-27
Melbourne, Australia	358	1296792	3622.3	7	16	-9
Osaka, Japan	342	1253204	3664.3	8	10	-2

M2 · Do High Follower Users Get More Engagement?

Challenge. Divide users into two groups: high follower users ($\geq 25,000$ followers) and low follower users ($< 25,000$ followers). Compare their average engagement per post.

cohort	n_users	n_posts	avg_engagement_per_post	total_engagement	sd_engagement
LOW ($< 25k$ followers)	758	5176	3647.98	18881951	1718.9
HIGH ($\geq 25k$ followers)	742	5045	3598.82	18156054	1729.1

Answer: 3598.82 average engagement per post for the high-follower cohort versus 3647.98 for the low-follower cohort — the lower-follower group is marginally ahead. The grain is the post, not the user, so a 17-post account does not outweigh a 2-post account.

M2b · The difference is noise. The -49.16 gap is -1.35% with $z = -1.441$, far inside ± 1.96 . Follower count does not predict engagement in this corpus — the same null result Round 1 found with a correlation of -0.0433, reached here by a completely different route. Two independent methods agreeing is worth more than either alone.

M3 · Most Active Users

Challenge. Find the top 10 users who have created the highest number of posts. Display their follower count, location, post count, and total engagement.

user_id	follower_count	location	n_posts	total_engagement
user_nfo3ih5u	40429	Barcelona, Spain	17	61730
user_n0ok02rt	2531	Dubai, UAE	16	49382
user_68enpikx	9364	Houston, USA	15	57816

user_id	follower_count	location	n_posts	total_engagement
user_xetn4exw	14902	Dubai, UAE	15	56209
user_8wk0j5ae	36943	Los Angeles, USA	15	51828
user_74ri5vbc	24934	New York, USA	15	49162
user_9mtets0p	8262	Johannesburg, South Africa	14	62414
user_gq0rypar	17672	Mumbai, India	14	60910

Answer: the ten busiest accounts, top of the list having written 17 posts. The order is total: post count, then total engagement, then user_id, so the tenth row cannot change between runs.

M3b · The cut is a tie band. 12 users share the 14-post count that sits on the boundary, out of only 17 distinct counts in the whole population. 'The top 10 most active users' is therefore one valid selection from a larger tied group — a fact the table cannot show on its own.

M4 · Platform Behaviour By High Follower Users

Challenge. Among users with at least 30,000 followers, determine which platform gives them the highest average engagement per post.

platform	n_posts	avg_engagement_per_post	rank_within_cohort
Instagram	684	3692.1	1
Twitter	674	3583.7	2
YouTube	726	3569.9	3
Reddit	675	3561.3	4
Facebook	699	3532.1	5

Answer: Instagram — 3,692.1 average engagement per post among accounts with at least 30,000 followers. The cohort filter is applied to the user before aggregation, which is what separates this from E3.

M4b · The winner survives every threshold. 30,000 is an arbitrary cut, so the same test is repeated at five of them. Instagram wins at all five, which makes the finding a property of the data rather than of the threshold:

follower_threshold	winning_platform	n_posts	avg_engagement_per_post
25000	Instagram	838	3658.3
30000	Instagram	684	3692.1
35000	Instagram	523	3689.7
40000	Instagram	366	3709.5
45000	Instagram	154	3706.1

M5 · Detect Suspicious Engagement

Challenge. Find posts where the number of shares is greater than the number of likes and comments combined. Such posts may represent unusual sharing behaviour. Return the top 20.

post_id	user_id	platform	likes	shares	comments	excess_shares	share_to_reaction_ratio
603lc32fqk41	user_bq4d19qm	Instagram	145	1965	15	1805	12.28
8df7c2foptde	user_vws4jlpy	Instagram	37	1915	118	1760	12.35
mtpfiec3whp4	user_mvu6hdsd	Unspecified	122	1880	92	1666	8.79
27w2t0eudwzl	user_cwh9v5dk	Twitter	6	1966	340	1620	5.68
ico83tj0cqa	user_rzirln2k	YouTube	61	1884	214	1609	6.85
agcvmdiq2joc	user_65x8ekq5	Instagram	91	1751	55	1605	11.99
m40jbon7sqsa	user_6wra58f7	Twitter	265	1915	83	1567	5.5
p52mh6x5bcle	user_m5o36yfr	Instagram	23	1608	21	1564	36.55

Answer: 1047 posts have more shares than likes and comments combined; the 20 largest excesses are shown, led by 603lc32fqk41 with 1,805 shares beyond its likes and comments.

M5b · The trap this challenge sets

Treating a missing like count as zero would report 2,201 suspicious posts instead of 1,047. 1,154 of those extra rows — 52.43% of the loose total — are anomalous only because a field is blank, and the reconciliation proves it: the arithmetic gap equals the missing-like count exactly. That is a missingness pattern being published as unusual sharing behaviour.

Hard challenges

H1 · Abnormally high engagement — and the honest empty answer

Challenge. Calculate the average total engagement per user. Identify users whose average engagement per post is more than twice the overall average engagement per post. Return the user ID, location, follower count, post count, and average engagement.

This query returns **zero rows**, and that is the correct answer. The overall average engagement per post is 3,623.72, so the 2x bar is 7,247.43. The highest per-post average any user achieves is 6,545.50 — 1.81x the baseline, about 19% short of the requirement. An empty result that is proved to be arithmetic is a finding; an empty result presented without that proof is indistinguishable from a broken query.

overall_avg_per_post	threshold_2x	users_examined	highest_user_avg	highest_avg_user	highest_avg_as_multiple_of_baseline	qualifying_users	verdict
3623.72	7247.43	1500	6545.5	user_2c5wzose	1.806	0	EMPTY BY ARITHMETIC -- no user reaches 2x, so H1 returning 0 rows is the answer

H1c · Where the question does have an answer

The same test at four thresholds turns a dead end into a calibration: the corpus has a tail, but that tail sits well below twice the mean.

multiple	threshold	users_qualifying
1.25x	4529.65	145
1.50x	5435.57	8
1.75x	6341.5	1
2.00x	7247.43	0

H2 · Top 3 users in every location

Challenge. For every location, rank users based on their total engagement. Return only the top 3 users from each location.

99 rows — 3 for each of 33 locations. ROW_NUMBER, not RANK, is what guarantees exactly three rows per location: RANK would return four whenever two users tie for third and would quietly break the 'only the top 3' instruction.

location	rank_in_location	user_id	follower_count	n_posts	total_engagement
Barcelona, Spain	1	user_nfo3ih5u	40429	17	61730
Barcelona, Spain	2	user_24wzfb8b	38633	14	53039
Barcelona, Spain	3	user_5s9ifp0y	28412	10	51089
Beijing, China	1	user_cdzp4vm2	11187	11	48019
Beijing, China	2	user_z6gqqt5r	13029	8	38518
Beijing, China	3	user_gre0h06x	23303	9	36116

H2b · Which podiums are actually contested

The tightest podium in the dataset is Tokyo, Japan, where third place (43,092) beats fourth (43,059) by 33 engagements — 0.08%. A ranking decided by 0.08% is decided by luck, and the companion query says so instead of letting the table imply otherwise.

location	third_place	fourth_place	absolute_margin	margin_pct
Tokyo, Japan	43092	43059	33	0.08
Dubai, UAE	49438	49382	56	0.11
Milan, Italy	42317	42255	62	0.15
Paris, France	39660	39550	110	0.28
New York, USA	41902	41684	218	0.52

H3 · Posts far above their own platform's average

Challenge. For every platform, identify posts whose engagement is significantly higher than the average engagement of that platform. A post is considered exceptional if its engagement is at least 2x the average engagement of its platform.

56 posts clear twice their own platform's mean. The benchmark is per-platform — a post on Twitter is judged against Twitter, not against the global mean — which is the entire point of the challenge. The bar itself is derived, not given, and differs by platform:

post_id	user_id	platform	likes	shares	comments	platform_avg_engagement	post_engagement	multiple_of_platform_avg
wo7py9aljg3t	user_o8le7hqf	Reddit	4864	1981	948	3606.9	7793	2.161
ycjj5zzt7mvx	user_d9971ba6	Instagram	4983	1919	991	3673.2	7893	2.149
5kvuyvf38nqx	user_z0feut2e	YouTube	4923	1971	861	3621.5	7755	2.141
gmoeib832zbs	user_pe5ycky	Facebook	4902	1880	982	3636.1	7764	2.135
a1kiwl618kzy	user_aaiari80	Facebook	4811	1952	920	3636.1	7683	2.113
pvfl3d8hj7jd	user_cs1uibwk	Instagram	4989	1840	909	3673.2	7738	2.107

platform	n_posts	platform_avg_engagement	exceptional_threshold	exceptional_posts	pct_of_platform	eligibility
Twitter	1742	3561.5	7123.0	16	0.92	labelled platform
Reddit	1716	3606.9	7213.8	14	0.82	labelled platform
YouTube	1770	3621.5	7243.0	11	0.62	labelled platform
Facebook	1763	3636.1	7272.3	8	0.45	labelled platform
Instagram	1689	3673.2	7346.3	7	0.41	labelled platform
Unspecified	1541	3646.9	7293.8	5	0.32	DATA GAP -- excluded from the H3 answer

H4 · Small accounts that out-engage the platform

Challenge. Find users who have fewer than 5,000 followers but whose total post engagement places them among the top 10% of all users. These users may represent unusually high performing or suspicious accounts. This requires multiple levels of analysis.

17 users have fewer than 5,000 followers yet rank in the top decile of total engagement — a four-level analysis: posts to users, users to deciles, decile 1, then re-joined to followership. NTILE splits tied totals arbitrarily at a boundary, so the same question is re-asked with an explicit percentile cut: both definitions return **17** users, and the decile floor is 38,765 engagement. The answer is a property of the data, not of the bucketing function.

user_id	location	follower_count	n_posts	total_engagement	avg_engagement_per_post	engagement_decile	top_decile_floor_engagement
user_uerv85na	Rome, Italy	1824	14	57044	4074.57	1	38765
user_hdas0iau	Rio de Janeiro, Brazil	1620	14	54678	3905.57	1	38765
user_fgjkkrie	Lyon, France	2211	12	53244	4437.0	1	38765
user_r7eg1rac	Houston, USA	2052	11	52606	4782.36	1	38765
user_n0ok02rt	Dubai, UAE	2531	16	49382	3086.38	1	38765
user_hvqqyzo	Melbourne, Australia	3213	12	45668	3805.67	1	38765
user_0aacyvtz	Rome, Italy	1452	11	45616	4146.91	1	38765
user_rr1uzkql	Milan, Italy	1069	13	44748	3442.15	1	38765

H5 · Corrupted posts — asked of the file that still has the corruption

Challenge. Identify potentially corrupted posts where one or more of the following conditions are true: (1) likes are negative; (2) platform is missing; (3) text content is missing; (4) text contains HTML entities/tags such as &, <div>, or
. Return the post ID and identify the type of anomaly.

4418 of the 12,000 distinct intake posts carry at least one defect, across the four families the challenge names. This question cannot be asked of the cleaned table: cleaning is exactly the step that removed the evidence, so a query over posts would return nothing and prove nothing. It runs against the verbatim intake staging table, and the defect predicates live in a SQL view so the definition of 'corrupted' is auditable in one place.

post_id	n_anomalies	anomaly_types
0oqpg2rgnqxa	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
26u0g2fea4nk	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
5b0pz3j0kcxl	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
6uafio2psiu	3	NEGATIVE_LIKES MISSING_PLATFORM HTML_ENTITY_OR_TAG
80rfm8lcs5rz	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
95tln4xx75o	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT

H5b · Inventory, reconciled against the Phase-1 corruption log

Each family is counted both ways: rows in the intake file and distinct posts, with the difference being the 360 exact replays. The intake counts — 525 negative likes, 1,846 missing platforms, 1,770 blank text bodies and 1,004 rows carrying HTML — are the same numbers Phase 1 recorded in output/anomaly_table.csv, computed independently in SQL. Two artefacts agreeing is the check that the recovery and the forensics describe the same file.

anomaly_type	intake_rows	distinct_posts	replay_rows_removed
MISSING_PLATFORM	1846	1784	62
MISSING_TEXT	1770	1711	59
HTML_ENTITY_OR_TAG	1004	974	30
NEGATIVE_LIKES	525	509	16

H5c · The posts carrying the most defects at once

20 posts carry three defects simultaneously. They are the rows most likely to break a naive pipeline and the reason the intake had to be staged verbatim rather than cleaned in place.

post_id	n_anomalies	anomaly_types
0oqpg2rgnqxa	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
26u0g2fea4nk	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
5b0pz3j0kcxl	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
6uafio2psiu	3	NEGATIVE_LIKES MISSING_PLATFORM HTML_ENTITY_OR_TAG
80rfm8lcs5rz	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT
95tln4xx75o	3	NEGATIVE_LIKES MISSING_PLATFORM MISSING_TEXT

H6 · Suspicious high-impact users

Challenge. Identify users who satisfy all three conditions: have fewer than 10,000 followers; their average post engagement is above the overall average; and at least one of their posts has more shares than likes. Rank these users by total engagement and display their location, follower count, number of posts, average engagement, and total engagement.

82 users satisfy all three conditions. The funnel shows the filters are selective, not accidental: 287 users have fewer than 10,000 followers, 748 beat the overall average, 1,045 have at least one over-shared post, and their intersection is 82. Condition three reuses the M5 rule, so 'more shares than likes' means the same thing in both places.

location	follower_count	n_posts	avg_engagement_per_post	total_engagement	engagement_rank	user_id
Johannesburg, South Africa	8262	14	4458.14	62414	1	user_9mtets0p
Houston, USA	9364	15	3854.4	57816	2	user_68enpikx
Rio de Janeiro, Brazil	1620	14	3905.57	54678	3	user_hdas0iau
Lyon, France	2211	12	4437.0	53244	4	user_fgjkkrie
Delhi, India	5153	11	4392.27	48315	5	user_zq2nib62

location	follower_count	n_posts	avg_engagement_per_post	total_engagement	engagement_rank	user_id
Vancouver, Canada	5971	10	4626.3	46263	6	user_acrn19py
Melbourne, Australia	3213	12	3805.67	45668	7	user_hvqqyzoi
Rome, Italy	1452	11	4146.91	45616	8	user_0aacyvtz

all_users	c1_fewer_than_10k_followers	c2_above_overall_avg	c3_has_over_shared_post	all_three_conditions
1500	287	748	1045	82

What this challenge set refused to publish

Six plausible findings were produced by the queries and then rejected by their own companion analysis. They are listed here because the rejections are the work: each one is a sentence that would have looked like insight in a slide deck.

- **"YouTube dominates the platform mix."** Rejected: $\chi^2 = 2.6094$ with $df = 4$ against a 9.488 critical value (E1b). The post counts are consistent with no platform preference at all.
- **"Instagram generates the most engagement."** Rejected as a finding: the lead is real as a mean but carries $z = 0.635$, so the ordering across platforms is indistinguishable from noise (E3b).
- **"Big accounts get more engagement."** Rejected: the high-follower cohort averages 3598.82 per post against 3647.98, $z = -1.441$ (M2b).
- **"These locations are engagement powerhouses."** Rejected: the total-engagement ranking is a post-volume ranking — the largest shift between the two orderings is 27 places (M1b).
- **"2,201 posts show abnormal sharing."** Rejected: 1,154 of those rows are only 'abnormal' because their like count is blank (M5b).
- **"Twice the average engagement identifies our power users."** Rejected: no user reaches it. The best is 1.81x, and the bar is only reachable at a lower multiple, where 145 users clear 1.25x and 8 clear 1.5x (H1b/H1c).

Reproduce it

`./run_all.sh` (or, step by step:)

```
python src/clean_data.py      # Phase 1 recovery, unchanged
python src/build_db.py        # schema + raw staging + anomaly scan
python src/run_sql.py         # Q1-Q12 and E/M/H, live capture
python src/make_screenshots.py
python -m pytest tests -q      # 42 invariants
python src/build_phase2_report.py
```

Artefacts: `queries/phase2_challenges.sql` · `output/phase2_sql_outputs.md` · `output/phase2_sql_results.json` · `output/screenshots/E1.png` ... `H6b.png`